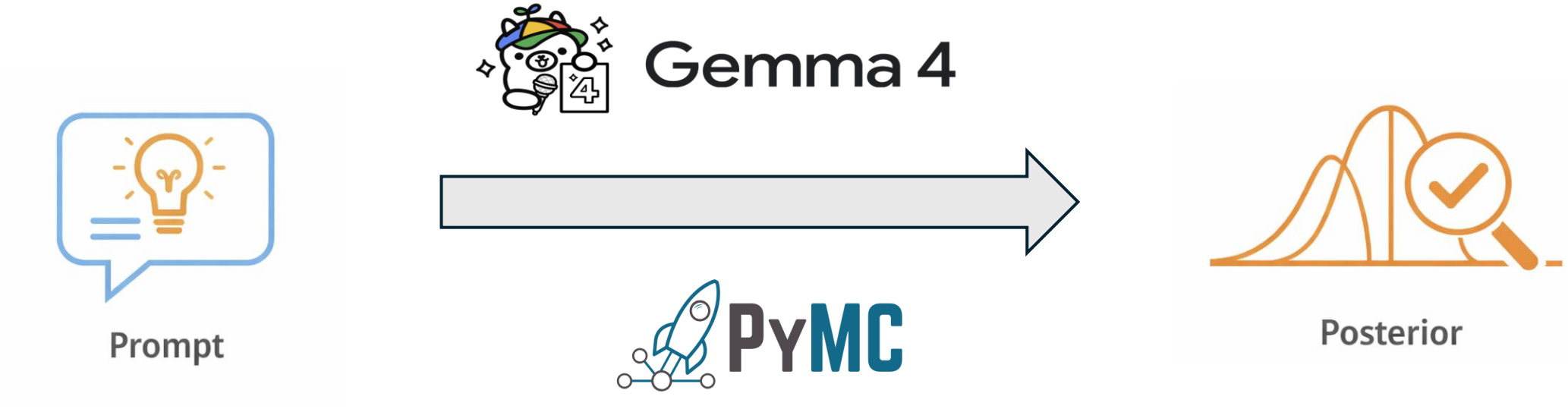

From Prompts to Posteriors: Agentic Bayesian A/B Testing with PyMC

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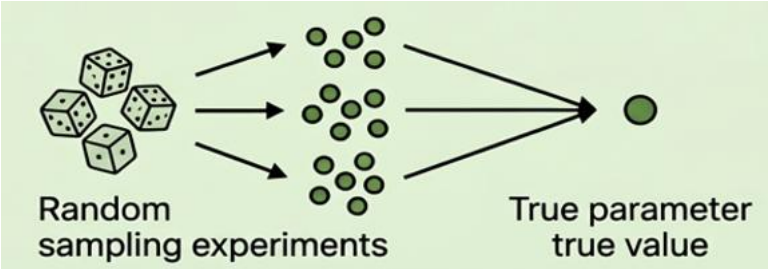
Image sources: ollama.com/library/gemma4, pymc.io and chatgpt prompt

Frequentist vs Bayesian

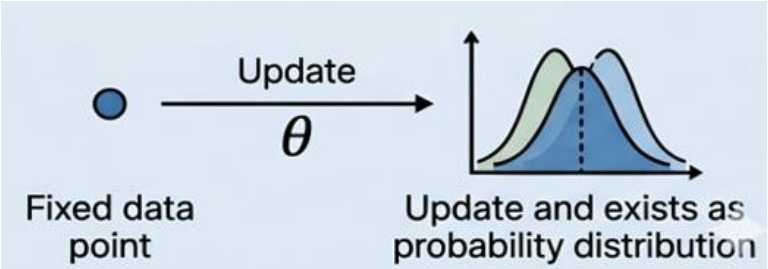
Frequentist View

- Data: Random
- Parameter : Fixed



Bayesian View

- Data: Fixed
- Parameter: Random



Data Source: Fonnesbeck & Wiecki — Probabilistic Programming and Bayesian Computing with PyMC | *PyData London 2024*

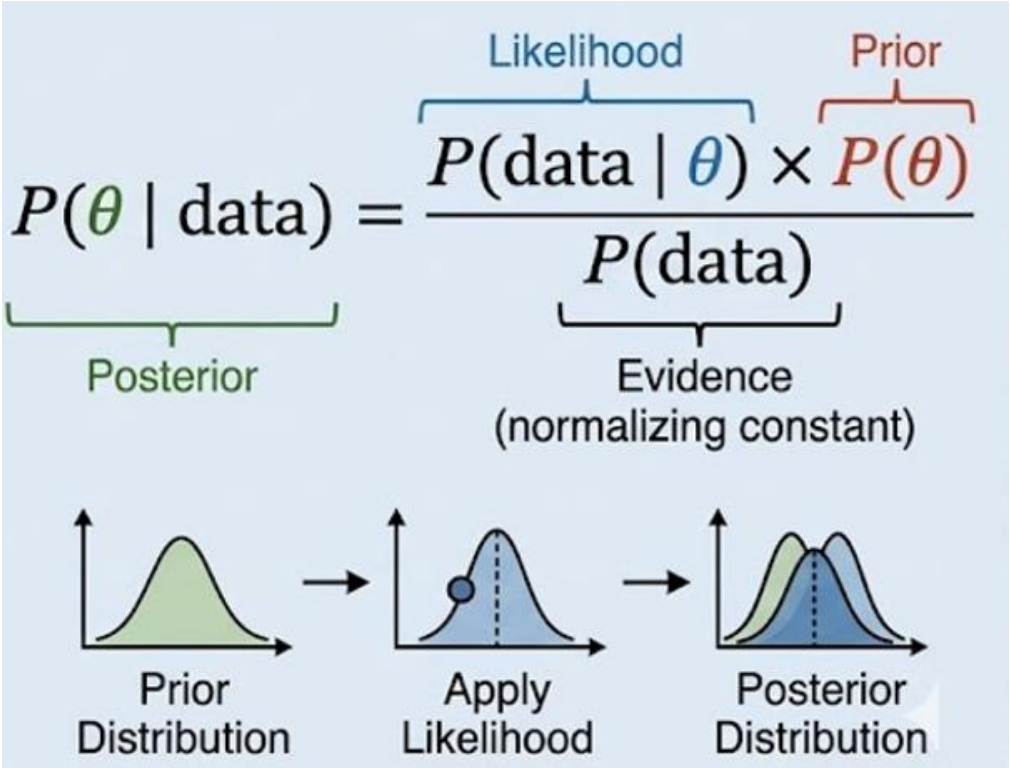
Images source: Gemini prompt

Ingredients of Bayesian Inference

Concepts

- **Prior P(θ)**
  - Belief about parameter before seeing data
- **Likelihood P(data| θ)**
  - How likely the data is given the parameter
- **Posterior P (θ | data)**
  - Updated belief after observing data

Bayes' Theorem



Data Source: Fonnesbeck & Wiecki — Probabilistic Programming and Bayesian Computing with PyMC | *PyData London 2024*
Image source: Gemini prompt

Bayesian A/B Testing

Experiment: Coffee shop tests two push notifications — which one gets more purchases?

A: *"Your morning coffee is waiting"* **B: *" Get 10% off your coffee today!"***

- **Prior:** Start with a belief (Conversion rate is 5-10% or we have no idea)
- **Likelihood:** Run experiment (A vs B) and collect data
- **Posterior:** Update your belief (conversion rate) using data

Use MCMC sampling (PyMC) to estimate probability B is better and by how much (lift)

Frequentist	Bayesian
p-value + confidence interval	P(B > A) + credible interval (HDI)
Reject / don't reject	94% chance B is better
❌ Indirect, not decision-focused	✅ Direct, intuitive, decision-ready

Decision: High probability + positive lift favors B

Agentic Bayesian A/B Testing

- Inspired by Andrej Karpathy's Autoresearch

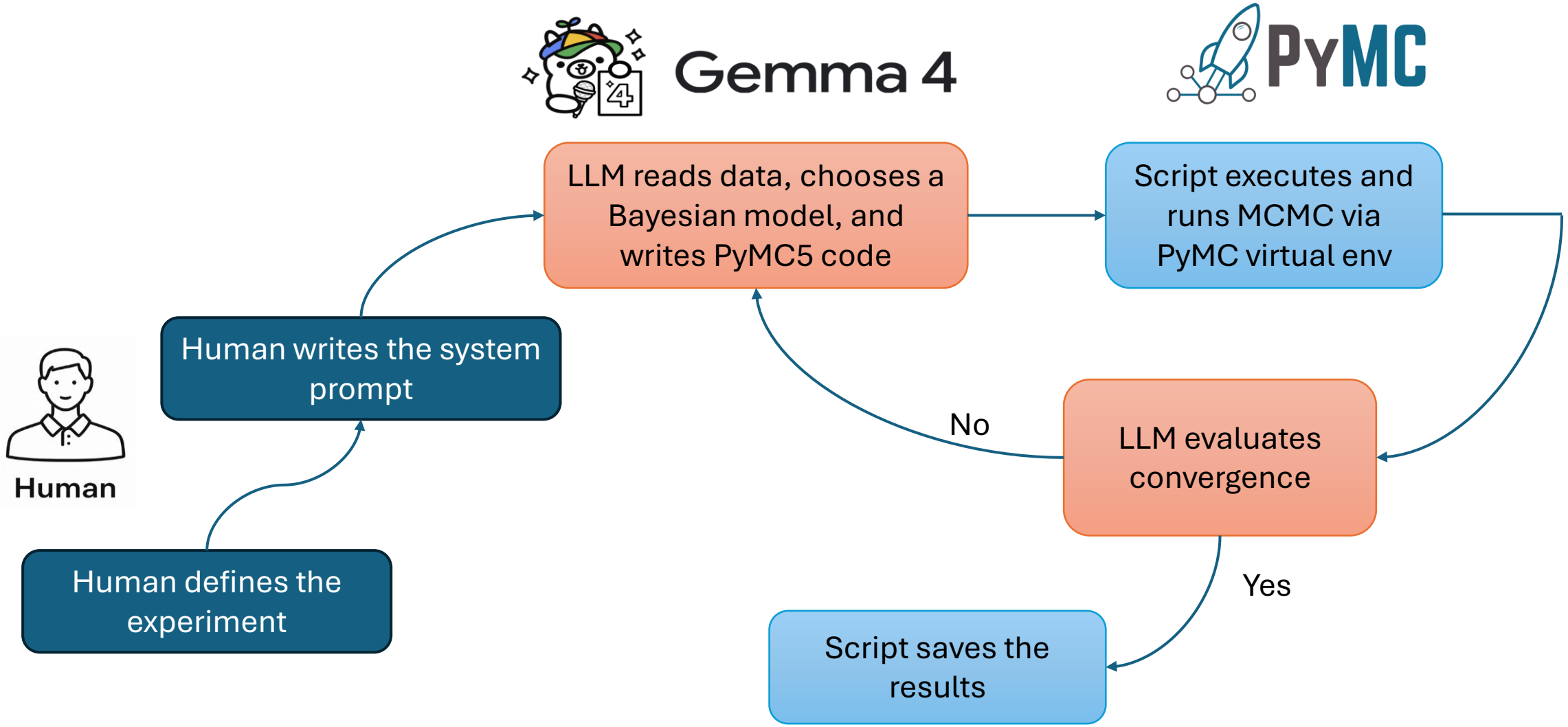


Image sources: ollama.com/library/gemma4 , pymc.io and chatgpt prompt

Results – Bayesian A/B Testing

A: "Your morning coffee is waiting"

Conversion Rate

Actual = 8.0%

Observed = 6.8%

B: "Get 10% off your coffee today!"

Conversion Rate

Actual = 13.0%

Observed = 13.4%

Lift

Actual = 62.5%

Observed = 97.1%

Weak Prior : Beta (1,1)

No prior knowledge

Lift = 99.6%

P (B >A) = 100.0%

Strong Prior: Beta(5,55)

Mean conversion rate – 8.3%

Lift = 89.0%

P (B >A) = 99.9%

Both Weak and Strong Priors Posterior MCMC chains converged

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Thank You